Mechanisms · TA Brief

Learning goal: I can identify levers, pulleys, and gears and explain how they let a small effort move a heavy load.

Watch (staff preview first)

Short silent teacher demonstration clip for this lesson's model and practical sequence · preview before class use.

Adult Support Routine

No cold calling; nobody is asked to read difficult text aloud.

Give time to start -> point to the relevant place -> rephrase -> offer a small prompt or word bank -> reduce to a limited choice.

Beside the You do in the pupil book, mark I (independent), S (supported) or C (choice).

Must-Know Words

mechanism: a simple machine such as a lever, pulley, or gear that changes a force

lever: a stiff bar that turns on a pivot to lift or move a load

pivot: the fixed resting point (fulcrum) that a lever turns around

pulley: a grooved wheel with a rope that changes the direction or size of a pull

Key Answers

A. Moving the effort end through a longer distance multiplies the force at the load end.

A. Meshed gears turn in opposite directions; driving a larger gear reduces speed and increases turning force.

1. The pivot (or fulcrum). 2. Close to the load and far from the effort. 3. They apply a smaller effort force over a longer distance (or more turns) to produce a larger force on the load over a shorter distance.